

Displays

D1:

$$\text{sd}(x, S) := \begin{cases} \inf_{y \in \partial S} \|y - x\|, & \text{if } x \notin S, \\ -\inf_{y \in \partial S} \|y - x\|, & \text{if } x \in S. \end{cases}$$

D2:

$$\dot{x} = F(x, u), \quad (1)$$

D3:

$$(u_0 u_1)(t) = \begin{cases} u_0(t), & t \in (0, a], \\ u_1(t), & t \in (a, a + b]. \end{cases}$$

D4:

$$\|F(y, u) - F(x, u)\| \leq L\|y - x\|, \quad \forall x, y \in S, \quad \forall u \in U.$$

D5:

$$\mathcal{R}_T(S) := \{x \mid \forall u \in \mathcal{U}^{(0, T]}, \exists t \in (0, T], \text{ s.t. } \phi(t, x, u) \in S\}.$$

D6:

$$\min\{D_t V(x, t) + H(x, t, \nabla V(x, t)), l(x) - V(x, t)\} = 0,$$

D7:

$$\mathcal{R}_T(\mathcal{X}_u) = \{x \mid V(x, T) \leq 0, x \in \mathcal{X}_u\},$$

D8:

$$\max_{u \in U} L_F h(x) + \kappa(h(x)) \geq 0, \quad (2)$$

D9:

$$k(x) \in \{u \in U \mid L_F h(x) + \kappa(h(x)) \geq 0\},$$

D10:

$$\exists t' > t \text{ with } \phi(t', x, u) \in S. \quad (3)$$

D11:

$$\exists t' > t, \text{ with } t' - t \in (0, \tau], \text{ and } \phi(t', x, u) \in S. \quad (4)$$

D12:

$$\max_{t \in (0, \tau]} e^{\gamma(h(\phi(t, x, u)))t} h(\phi(t, x, u)) \geq h(x), \quad (5)$$

D13:

$$\gamma_{\alpha, \beta}(s) = \begin{cases} \alpha, & \text{if } s \geq 0, \\ \beta, & \text{if } s < 0, \end{cases} \quad (6)$$

D14:

$$\tau_0 := \max \left\{ \arg \max_{t \in (0, \tau]} e^{\gamma(h(\phi(t, x_0, u)))t} h(\phi(t, x_0, u)) \right\}, \quad (7)$$

D15:

$$u_{[n]} = u_0|_{(0, \tau_0]} u_1|_{(0, \tau_1]} \cdots u_n|_{(0, \tau_n]} \in \mathcal{U}^{(0, t_n]}$$

D16:

$$x_{n+1} = \phi(\tau_n, x_n, u_n|_{(0, \tau_n]}) = \phi(t_n, x, u), \quad (8)$$

D17:

$$h(\phi(t'' + t, x, u)) < 0 \quad \forall t \in (0, \tau],$$

D18:

$$\liminf_{t \rightarrow \infty} d(h_{\geq 0}, \phi(t, x, u)) = 0,$$

D19:

$$(h(x) - a_1 \text{sd}(x, h_{\leq 0}))(h(x) - a_2 \text{sd}(x, h_{\leq 0})) \leq 0 \quad (9)$$

D20:

$$\kappa_{\alpha, \beta}(s) := \gamma_{\alpha, \beta}(s) s. \quad (10)$$

D21:

$$\hat{h}(\cdot) = -\text{sd}(\cdot, S)$$

D22:

$$\max_{t \in (0, \hat{\tau}]} e^{\hat{\gamma}(\hat{h}(\phi(t, x, u)))t} \hat{h}(\phi(t, x, u)) \geq \hat{h}(x) \quad (11)$$

D23:

$$\begin{aligned} \bar{\delta} &:= \sup_{x \in D_0} (\text{sd}(x, S) - \text{sd}(x, h_{\geq 0})), \\ \underline{\delta} &:= \inf_{x \in D_0} (\text{sd}(x, S) - \text{sd}(x, h_{\geq 0})). \end{aligned}$$

D24:

$$|\text{sd}(\phi(t, y, u), S) - \text{sd}(\phi(t, x, u), S)| \leq re^{Lt}, \forall t \geq 0, \quad (12)$$

D25:

$$\forall t \in [0, \tau], \text{sd}(\phi(t, x, u), \mathcal{X}_u) > re^{Lt}, \quad (13)$$

D26:

$$\exists t \in (0, \tau], \text{ s.t. } \text{sd}(\phi(t, x, u), \mathcal{X}_u) < -re^{Lt}, \quad (14)$$

D27:

$$\text{sd}(\phi(t, y, u), \mathcal{X}_u) \geq \text{sd}(\phi(t, x, u), \mathcal{X}_u) - re^{-Lt} > 0,$$

D28:

$$\mathcal{B}_r(x) \cap \mathcal{R}_\tau(\mathcal{X}_u) = \emptyset.$$

D29:

$$\text{sd}(\phi(t^*, x, u), \mathcal{X}_u) < -re^{Lt^*}.$$

D30:

$$\text{sd}(\phi(t^*, y, u), \mathcal{X}_u) \leq \text{sd}(\phi(t^*, x, u), \mathcal{X}_u) + re^{Lt^*} < 0,$$

D31:

$$\mathcal{B}_r(x) \subseteq R_{t^*}(\mathcal{X}_u).$$

D32:

$$\hat{h}_r^-(x, u, t) := h(\phi(t, x, u)) - re^{Lt} \quad (15)$$

D33:

$$\max_{t \in (0, \tau]} e^{\gamma(\hat{h}_r^-(x, u, t))t} \hat{h}_r^-(x, u, t) \geq h(x) + r, \quad (16)$$

D34:

$$\max_{t \in (0, \tau]} e^{\gamma(h(y, u, t))t} h(\phi(t, y, u)) \geq h(y), \quad (17)$$

D35:

$$\hat{h}_r^+(x, u, t) := h(\phi(t, x, u)) + re^{Lt} \quad (18)$$

D36:

$$\max_{t \in (0, \tau]} e^{\gamma(\hat{h}_r^+(x, u, t))t} \hat{h}_r^+(x, u, t) < h(x) - r \quad (19)$$

D37:

$$\max_{t \in (0, \tau]} e^{\gamma(h(\phi(t, y, u)))t} h(\phi(t, y, u)) < h(y). \quad (20)$$

D38:

$$\begin{aligned} t^* &= \arg \max_{t \in (0, \tau]} \max_{u \in \mathcal{U}^{(0, \tau]}} e^{\gamma(\hat{h}_r^-(x, u, t))t} \hat{h}_r^-(x, u, t), \\ u^* &= \arg \max_{u \in \mathcal{U}^{(0, \tau]}} \max_{t \in (0, \tau]} e^{\gamma(\hat{h}_r^-(x, u, t))t} \hat{h}_r^-(x, u, t). \end{aligned}$$

D39:

$$\begin{aligned} &\max_{u \in \mathcal{U}, t \in (0, \tau]} e^{\gamma(h(y, u, t))t} h(\phi(t, y, u)) \\ &\geq e^{\gamma(h(y, u^*, t^*))t^*} h(y, u^*, t^*) \\ &\geq e^{\gamma(\hat{h}_r^-(x, u, t^*))t^*} \hat{h}_r^-(x, u, t^*) \\ &\geq h(x) + r \\ &\geq h(y), \end{aligned}$$

D40:

$$t^* = \arg \max_{t \in (0, \tau]} \max_{u \in \mathcal{U}^{(0, \tau]}} e^{\gamma(h(y, u, t))t} h(y, u, t),$$

$$u^* = \arg \max_{u \in \mathcal{U}^{(0, \tau]}} \max_{t \in (0, \tau]} e^{\gamma(h(y, u, t))t} h(y, u, t).$$

D41:

$$\begin{aligned} & \max_{u \in \mathcal{U}^{(0, \tau]}, t \in (0, \tau]} e^{\gamma(h(y, u, t))t} h(y, u, t) \\ & \leq e^{\gamma(\hat{h}_r^+(x, u^*, t^*))t^*} \hat{h}_r^+(x, u^*, t^*) \\ & \leq \max_{u \in \mathcal{U}^{(0, \tau]}, t \in (0, \tau]} e^{\gamma(\hat{h}_r^+(x, u, t))t} \hat{h}_r^+(x, u, t) \\ & < h(x) - r \\ & < h(y), \end{aligned}$$

D42:

$$\dot{x} = \frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -v + v \cos x_3 + ux_2 \\ v \sin x_3 - ux_1 \\ -u \end{bmatrix},$$

D43:

$$\begin{aligned} \|F(x, u) - F(y, u)\| &\leq L\|x - y\| \\ \|F(x, u) - F(x, v)\| &= \|g(x)(u - v)\| \leq L_u\|u - v\| \end{aligned}$$

D44:

$$\begin{aligned} & \|\phi(t, x, u) - \phi(t, y, u)\| \\ & \leq e^{Lt}\|x - y\| + L_u \int_0^t e^{L(t-s)} \|u(s) - u(s)\| ds \\ & = e^{Lt}\|x - y\| \\ & \leq re^{Lt}, \end{aligned}$$

D45:

$$\begin{aligned} & |\text{sd}(\phi(t, x, u), S) - \text{sd}(\phi(t, y, u), S)| \\ & = \|\phi(t, x, u) - x^*\| - \|\phi(t, y, u) - y^*\| \\ & \leq \|\phi(t, x, u) - x^*\| - \|\phi(t, y, u) - x^*\| \\ & \leq \|\phi(t, x, u) - \phi(t, y, u)\| \\ & \leq re^{Lt}, \end{aligned}$$

D46:

$$\begin{aligned} & |\text{sd}(\phi(t, x, u), S) - \text{sd}(\phi(t, y, u), S)| \\ & = \|\phi(t, x, u) - x^*\| - \|\phi(t, y, u) - y^*\| \\ & \leq \|\phi(t, x, u) - x^*\| - \|\phi(t, y, u) - x^*\| \\ & \leq \|\phi(t, x, u) - \phi(t, y, u)\| \\ & \leq re^{Lt}, \end{aligned}$$

D47:

$$\begin{aligned} & |\text{sd}(\phi(t, x, u), S) - \text{sd}(\phi(t, y, u), S)| \\ & = \|\phi(t, y, u) - y^*\| + \|\phi(t, x, u) - x^*\| \\ & \leq \|\phi(t, x, u) - p^*\| + \|\phi(t, y, u) - p^*\| \\ & = \|\phi(t, x, u) - \phi(t, y, u)\| \\ & \leq re^{Lt}, \end{aligned}$$

Inline

I1: τ

I2: $\|\cdot\|$

I3: $\mathbb{R}^n$

I4: $x \in \mathbb{R}^n$
 I5: $r > 0$
 I6: r
 I7: x
 I8: $\mathcal{B}_r(x) := \{y \in \mathbb{R}^n \mid \|y - x\| \leq r\}$
 I9: $S \subseteq \mathbb{R}^n$
 I10: S
 I11: $x \in \mathcal{X} \subseteq \mathbb{R}^n$
 I12: $\mathcal{X}$
 I13: $u \in U \subseteq \mathbb{R}^m$
 I14: $\mathcal{U}^{(a,b]} := \{u : (a, b] \rightarrow U \mid u \text{ is measurable}\}$
 I15: $(a, b]$
 I16: $\mathcal{U} := \mathcal{U}^{(0, +\infty)}$
 I17: $u_0 \in \mathcal{U}^{(0, a]}$
 I18: $u_1 \in \mathcal{U}^{(0, b]}$
 I19: $u_0 u_1 \in \mathcal{U}^{(0, a+b]}$
 I20: $u \in \mathcal{U}^{(a, b]}$
 I21: $(c, d] \subset (a, b]$
 I22: $u|_{(c, d]}$
 I23: u
 I24: $(c, d]$
 I25: $u_n \in \mathcal{U}^{(0, \tau_n]}$
 I26: $\tau_n > 0$
 I27: $n \in \mathbb{N}$
 I28: $u_{[n]} := u_0 u_1 \cdots u_n$
 I29: $u_{[\infty]} := \lim_{n \rightarrow \infty} u_{[n]}$
 I30: U
 I31: $\mathcal{U}^{(a, b]}$
 I32: $u \in \mathcal{U}^{(0, a]}$
 I33: $\phi(t, x, u)$
 I34: $t \in (0, a]$
 I35: $u \in \mathcal{U}$
 I36: $\phi(\cdot, x, u)$
 I37: $[0, \infty)$
 I38: $F(x, u)$
 I39: $L \geq 0$
 I40: $u(\cdot) \in \mathcal{U}$
 I41: $\mathcal{X}_u \subset \mathcal{X}$
 I42: $t \geq 0$
 I43: $\mathcal{X} \setminus \mathcal{X}_u$
 I44: $\phi(t, x, u)$
 I45: $x \in \mathcal{X}$
 I46: $\mathcal{X}_u$
 I47: $\exists u \in \mathcal{U}$
 I48: $\forall t \geq 0,$
 I49: $\phi(t, x, u) \notin \mathcal{X}_u$
 I50: $\mathcal{C}$
 I51: $\mathcal{X}_u$
 I52: $\mathcal{C} \subseteq \mathcal{X}$
 I53: $x \in \mathcal{C}$
 I54: $\phi(t, x, u) \in \mathcal{C}$
 I55: $t \geq 0$
 I56: $T > 0$
 I57: T
 I58: $T = \infty$
 I59: $\mathcal{R}(S)$

I60: $V(x, t) := \min_{s \in [-t, 0]} l(\phi(s, x, u))$
 I61: $V(x, t)$
 I62: $l(x)$
 I63: $V(x, 0) = l(x)$
 I64: $H(x, t, \nabla V(x, t)) := \max_{u \in U} D_x V(x, t) \cdot f(x, u)$
 I65: $V(x, T)$
 I66: $u(\cdot)$
 I67: $x \in \mathcal{R}_{+\infty}^c(\mathcal{X}_u)$
 I68: $\dot{h}$
 I69: $\mathcal{K}$
 I70: $\kappa : \mathbb{R} \rightarrow \mathbb{R}$
 I71: $\kappa(0) = 0$
 I72: $h(x)$
 I73: κ
 I74: $L_F h(x) = \frac{\partial h}{\partial x}^\top F(x, u)$
 I75: $k(x)$
 I76: $h_{\geq 0} := \{x : h(x) \geq 0\}$
 I77: $h_{\geq 0}$
 I78: h
 I79: $h_{\geq 0} \cap \mathcal{X}_u = \emptyset$
 I80: $h_{\geq 0}$
 I81: $x \in h_{\geq 0}$
 I82: $\phi(t, x, u) \in h_{\geq 0}, \forall t > 0,$
 I83: $h_{\geq 0}$
 I84: $x \in S$
 I85: $\exists$
 I86: $u \in \mathcal{U}$
 I87: $t \geq 0$
 I88: $\tau > 0$
 I89: ∂S
 I90: $h : \mathbb{R}^n \rightarrow \mathbb{R}$
 I91: $x \in D_0 := h_{\geq -c}$
 I92: $c > 0$
 I93: $\exists u \in \mathcal{U}^{(0, \tau]}$
 I94: $\gamma : \mathbb{R} \rightarrow \mathbb{R}_{>0}$
 I95: $\sup$
 I96: $(0, \tau]$
 I97: $\max$
 I98: $-\infty$
 I99: γ
 I100: α
 I101: β
 I102: $h_{\geq 0} \cap \mathcal{R}_\tau(\mathcal{X}_u) = \emptyset$
 I103: $\phi(t, x, u) \notin \mathcal{X}_u$
 I104: $t \geq 0$
 I105: $x \in h_{\geq 0}$
 I106: $x_0 := x$
 I107: $t_0 := 0$
 I108: $u_0 \in \mathcal{U}^{(0, \tau]}$
 I109: $x_1 := \phi(\tau_0, x_0, u_0) \in h_{\geq 0}$
 I110: $t_1 := \tau_0 + t_0$
 I111: $x_n \in h_{\geq 0}$
 I112: t_n
 I113: $u_n \in \mathcal{U}^{(0, \tau]}$
 I114: $\tau_n \in (0, \tau]$
 I115: $x_{n+1} := \phi(\tau_n, x_n, u_n) \in h_{\geq 0}$

I116: $t_{n+1} = \tau_n + t_n$
 I117: u_n
 I118: $(0, \tau_n]$
 I119: $u = \lim_{n \rightarrow \infty} u_{[n]} \in \mathcal{U}^{(0, t^*)}$
 I120: $t^* = \lim_{n \rightarrow \infty} t_n$
 I121: $t^* = \infty$
 I122: $n \geq 0$
 I123: $\phi(t_n, x, u) \in h_{\geq 0}$
 I124: $n \geq 0$
 I125: $t_{n+1} - t_n \in (0, \tau]$
 I126: $t_n \rightarrow \infty$
 I127: $x \in h_{\geq 0}$
 I128: (i)
 I129: $h_{\geq 0} \cap \mathcal{R}_\tau(\mathcal{X}_u) = \emptyset$
 I130: $u \in \mathcal{U}$
 I131: $t' > 0$
 I132: $\phi(t', x, u) \in \mathcal{X}_u$
 I133: $h(x) \geq 0$
 I134: $h < 0$
 I135: $t'' \in [0, t']$
 I136: $h(\phi(t'', x, u)) = 0$
 I137: $h(\phi(t, x, u)) < 0$
 I138: $t \in (t'', t']$
 I139: $\phi(t'', x, u)$
 I140: $t' - t'' > \tau$
 I141: $\square$
 I142: $x \in D_0 \setminus h_{\geq 0}$
 I143: $d(S, x) := \min_{y \in S} \|y - x\|$
 I144: $h : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$
 I145: $\exists a_1, a_2 > 0$
 I146: $x \in D$
 I147: $D_0 := h_{\geq -c}$
 I148: $a_2 > a_1 > 0$
 I149: $\kappa_{\alpha, \beta}$
 I150: $\alpha > 0$
 I151: $\beta > 0$
 I152: $h_{\geq 0} \subseteq S \subseteq h_{\geq -c}$
 I153: $\partial S \cap h_{=0} = \emptyset$
 I154: $\hat{D}_0 := \hat{h}_{\geq -\hat{c}}$
 I155: $\hat{c} \geq 0$
 I156: $\hat{h}_{\geq -\hat{c}} \subseteq h_{\geq -c}$
 I157: $x \in \hat{h}_{\geq -\hat{c}}$
 I158: $u \in \mathcal{U}^{(0, \tau]}$
 I159: $\hat{\gamma} := \gamma_{\hat{\alpha}, \hat{\beta}}$
 I160: $\hat{\alpha}, \hat{\beta} > 0$
 I161: $\hat{\alpha} > \alpha, \hat{\beta} < \beta$
 I162: $\hat{\tau} \geq \max\{\frac{\log(a_2/a_1)}{\hat{\alpha}-\alpha}, \frac{\log(a_2/a_1)}{\beta-\hat{\beta}}\} + \frac{\log(\bar{\delta}/\underline{\delta})}{\min\{\hat{\alpha}, \hat{\beta}\}}$
 I163: $\mathcal{B}_r(x)$
 I164: $\phi(t, y, u)$
 I165: y
 I166: $\|x - y\| \leq r$
 I167: L
 I168: $\mathcal{B}_r(\cdot)$
 I169: $\mathcal{R}_\tau(\mathcal{X}_u)$
 I170: $\mathcal{R}_\tau(\mathcal{X}_u)$
 I171: $\mathcal{B}_r(x) := \{y \mid \|y - x\| \leq r\}$

I172: $x \in \mathcal{X}$
 I173: $u \in \mathcal{U}^{(0,\tau]}$
 I174: $\mathcal{B}_r(x) \cap \mathcal{R}_\tau(\mathcal{X}_u) = \emptyset$
 I175: $x \in \mathcal{X}$
 I176: $u \in \mathcal{U}^{(0,\tau]}$
 I177: $\mathcal{B}_r(x) \subseteq \mathcal{R}_\tau(\mathcal{X}_u)$
 I178: $t \in [0, \tau]$
 I179: $y \in \mathcal{B}_r(x)$
 I180: $t^* < \tau$
 I181: $h(\cdot) := -\text{sd}(\cdot, S)$
 I182: $\gamma := \gamma_{\alpha,\beta}$
 I183: $\alpha, \beta > 0$
 I184: $\exists u \in \mathcal{U}^{(0,\tau]}$
 I185: $\forall u \in \mathcal{U}^{(0,\tau]}$
 I186: t^*
 I187: u^*
 I188: $t^* \in (0, \tau]$
 I189: $S \subset \mathcal{X}$
 I190: $h = -\text{sd}(x, S)$
 I191: $\mathcal{G} := \{g_i := \mathcal{B}_{r_i}(x_i)\}_{i=1}^{|\mathcal{G}|}$
 I192: $g_i \cap g_j = \emptyset, i \neq j$
 I193: $\cup \mathcal{G} := \cup_{i=1}^{|\mathcal{G}|} g_i = \mathcal{X}$
 I194: $\mathcal{G}_s$
 I195: $\mathcal{G}_u$
 I196: $\mathcal{X} = (\cup \mathcal{G}_u) \cup (\cup \mathcal{G}_s)$
 I197: $S = \cup \mathcal{G}_s$
 I198: $2, 3,$
 I199: 4
 I200: $1, 2,$
 I201: 3
 I202: $\mathcal{X}, \tau$
 I203: $\mathcal{G}_s, \mathcal{G}_u = \text{VerifyCells}(\mathcal{X}, \mathcal{X}_u, 0, (13), (14))$
 I204: $\mathcal{G}_s, \mathcal{G}_u = \text{VerifyCells}(\mathcal{G}_s, \mathcal{G}_u, \tau, (13), (14))$
 I205: $\mathcal{G}_s, \mathcal{G}_u = \text{VerifyCells}(\mathcal{G}_s, \mathcal{G}_u, \tau, (17), (20))$
 I206: $\triangleright$
 I207: $\mathcal{G}_s$
 I208: $\mathcal{G}_u$
 I209: $S \subseteq \mathcal{X}$
 I210: $\mathcal{R}_\tau(\cup \mathcal{G}_u)$
 I211: $S = \cup \mathcal{G}_s$
 I212: $\mathcal{C}_s$
 I213: $\mathcal{C}_u$
 I214: $\mathcal{G}, \mathcal{G}_u, \tau, \mathcal{C}_s, \mathcal{C}_u$
 I215: $\mathcal{G}$
 I216: $\tau \geq 0$
 I217: $\mathcal{C}_s$
 I218: $\mathcal{C}_u$
 I219: $\mathcal{G} \neq \emptyset$
 I220: $\forall g_i = \mathcal{B}_r(x) \in \mathcal{G}, i \in \mathbb{N}$
 I221: $\mathcal{G} \leftarrow \mathcal{G} - \{g_i\}$
 I222: $\text{SafetyCheck}(g_i, \mathcal{G}, \mathcal{G}_u, \tau, \mathcal{C}_s, \mathcal{C}_u)$
 I223: $\mathcal{C}_s$
 I224: $\mathcal{C}_u$
 I225: SafetyCheck
 I226: $g_i, \mathcal{G}, \mathcal{G}_u, \tau, \mathcal{C}_s, \mathcal{C}_u$
 I227: g_i

I228: $\mathcal{G}, \mathcal{G}_s, \mathcal{G}_u$
 I229: $\mathcal{X}_u = \mathcal{G}_u$
 I230: $\mathcal{G}_s = \emptyset$
 I231: n_s
 I232: $\mathcal{G}_u \leftarrow \mathcal{G}_u \cup \{g_i\}$
 I233: $\mathcal{G}_s \leftarrow \mathcal{G}_s \cup \{g_i\}$
 I234: $\mathcal{G} \leftarrow \mathcal{G} \cup \text{SplitCell}(g_i)$
 I235: g
 I236: $g = \mathcal{B}_r(x) \in \mathcal{G}$
 I237: $(x_1, x_2, \dots, x_n) = x$
 I238: $P := \{x + \frac{2r}{3} \cdot \delta \mid \delta \in \{-1, 0, 1\}^n\}$
 I239: $\mathcal{G}_{split} := \{\mathcal{B}_{\frac{r}{3}}(p) \mid p \in P\}$
 I240: $[x_1, x_2]^T \in \mathbb{R}^2$
 I241: $x_3 \in [0, 2\pi]$
 I242: $v \geq 0$
 I243: $u \in [-1, 1]$
 I244: $\sqrt{x_1^2 + x_2^2} \leq 1$
 I245: x_3
 I246: $\tau = 1$
 I247: $n_s = 3000$
 I248: V_{BRT}
 I249: $r_{\min} = 0.041$
 I250: $(V_{BRT \cap h' \leq 0} / V_{BRT})$
 I251: $\beta = \alpha = 0.05$
 I252: $\alpha = 1$
 I253: $n_s = 3000$
 I254: $x_3 = \pi$
 I255: $\tau \rightarrow 0$
 I256: $(V_\tau - V_{\text{BRT}}) / V_{\text{BRT}}$
 I257: t
 I258: $n_s = 3000$
 I259: $r_{\min} = 0.370$
 I260: $L_u = \max_{t \in (0, \tau]} \|g(\phi(t, x, u))\| := \max_{\|v\|=1, t \in (0, \tau]} \|g(\phi(t, x, u))v\| > 0$
 I261: x'
 I262: y'
 I263: $\phi(t, y, u)$
 I264: $\forall t \in (0, \tau]$
 I265: $x^* = \arg \min_{x^* \in \partial S} \text{sd}(\phi(t, x, u), \mathcal{S}), y^* = \arg \min_{y^* \in \partial S} \text{sd}(\phi(t, y, u), \mathcal{S})$
 I266: $\lambda \in [0, 1]$
 I267: $\lambda \phi(t, x, u) + (1 - \lambda) \phi(t, y, u) \in \partial S$
 I268: $p^* := \lambda \phi(t, x, u) + (1 - \lambda) \phi(t, y, u) \in \partial S$
 I269: p^*
 I270: $|\text{sd}(\phi(t, x, u), S) - \text{sd}(\phi(t, y, u), S)| \leq r e^{Lt}$